

Progress Report – Phase 4

Project: Industry AI Flow – AI-Powered Construction Industry Platform

Program: Integrated Artificial Intelligence, SAIT Capstone Project

Team Members: - Angel Daniel Bustamante Perez - Jason Niu - Jack Si

Instructor: Reeta

Date: March 2026

Project Status Summary

Phase 4 focuses on the machine learning development process using the CRISP-DM methodology. We built, trained, and evaluated multiple regression models on our 10,000-record construction cost dataset. The system is feature-complete and we are preparing for the Capstone Showcase.

Task Distribution

Jason Niu – Software Development Lead

Task	Status	Description
Jupyter notebook development	Complete	Full CRISP-DM notebook with EDA, data prep, model building, and evaluation
Data preprocessing pipeline	Complete	Feature engineering, standardization, one-hot encoding, train/test split
Model training (6 algorithms)	Complete	Ridge, Lasso, ElasticNet, Random Forest, Gradient Boosting, SVR
Hyperparameter tuning	Complete	GridSearchCV on Gradient Boosting and Ridge Regression
Model evaluation & visualization	Complete	Performance comparison charts, residual analysis, feature importance
Production model integration	Complete	Ridge Regression deployed as FastAPI endpoint with confidence intervals
System testing & optimization	Complete	516+ unit tests, performance optimization, security hardening

Jack Si – Construction Domain Expert

Task	Status	Description
Dataset curation	Complete	Provided 10,000-row construction project dataset with 25 features

Task	Status	Description
Feature identification	Complete	Identified key cost drivers from construction industry knowledge
Business understanding documentation	Complete	Defined problem context, objectives, and business impact
Model output validation	Complete	Verified prediction results against industry experience
Demo scenario preparation	In Progress	Preparing realistic construction project scenarios for showcase

Angel Daniel Bustamante Perez – Project Support

Task	Status	Description
Project coordination	Complete	Team communication and milestone tracking
Document preparation	In Progress	Assignment submissions and presentation materials
Testing support	Complete	Assisted with system testing and bug reporting
Presentation preparation	In Progress	Preparing Capstone Showcase presentation slides

Milestones

Milestone	Target Date	Status
Phase 1: Project proposal	Completed	Done
Phase 2: Data collection & preprocessing	Completed	Done
Phase 3: System architecture design	Completed	Done
Phase 4: Model development (CRISP-DM)	March 2026	Done
Full system integration	Completed	Done
36 rounds of test-driven improvement	Completed	Done
Demo preparation & rehearsal	Late March 2026	In Progress
Capstone Showcase	Late March / Early April 2026	Upcoming

Challenges and Solutions

Challenge	Solution
Selecting the right ML algorithm	We trained and compared 6 different algorithms to find the best fit
Avoiding data leakage	We excluded post-hoc outcome columns (actual_cost, actual_duration) from features
Balancing model accuracy vs. interpretability	We chose Ridge Regression for production (interpretable) while noting Gradient Boosting achieves higher R2
Feature engineering for categorical variables	Applied one-hot encoding for project_type (12 categories) and location (15 cities)
Ensuring model generalization	Used 5-fold cross-validation and verified train/test metrics are consistent (no overfitting)

Next Steps

1. Final demo rehearsal with realistic construction project scenarios
2. Prepare Capstone Showcase presentation slides
3. Ensure system stability during live demo
4. Complete final assignment submissions